

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Rearranging Formulas

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

REARRANGING FORMULAS · CH₂ · L7



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The Map

This lesson collapses Rearranging Formulas into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Make the subject visible

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Clear fractions first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Undo operations in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Factor out the subject

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Rearrange formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Square both sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 19 website questions, 19 checked solutions, 3 local PDFs read.

01 Make the subject visible

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$d = g + 2ac$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q3

(a)

$$d = g + 2ac$$

(b)

$$9ef - 12f = 3f(3e - 4)$$

Finish: $a = \frac{d-g}{2c}, 3f(3e-4), x^2 - 3x - 10, n^6$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

3 (a) Make a the subject of $d = g + 2ac$ (2) (b) Factorise fully $9e^2f - 12ef^2$ (2) (c) Expand and simplify $(x+2)(x-5)$ (2) (d) Simplify fully $n^4n^7n^5$ (2)

YOUR SOLUTION

02 Clear fractions first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q14

(a)

$$\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$$

Multiply by 20

$$4(9a - 7) - 5(3a - 7) = 91$$

Finish: (a) $a = 4$, (b) $c = \frac{8-3p^2}{p^2-a}$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

14 (a) Solve $9.75374a - 4.55 = 4.55$ Show clear algebraic working. $a = (3)$ (b) Make c the subject of the formula $p = ac + 83$ (4)

YOUR SOLUTION

03 Undo operations in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{c}{y} - 5z$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q6

(a)

$$A = \frac{c}{y} - 5z$$

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $c = y(A + 5z)$, 1, $(x - 3)(x - 8)$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 (a) Make c the subject of $A = c y - 5z$ (2) (b) Write down the value of g 0 (1) (c) Factorise $x^2 - 11x + 24$ (2)

YOUR SOLUTION

04 Factor out the subject

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$d = g + 2ac$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q3

(a)

$$d = g + 2ac$$

(b)

$$9ef - 12f = 3f(3e - 4)$$

Finish: $a = \frac{d-g}{2c}, 3f(3e-4), x^2 - 3x - 10, n^6$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

- 3 (a) Make a the subject of $d = g + 2ac$ (2) (b) Factorise fully $9e^2f - 12ef^2$ (2) (c) Expand and simplify $(x + 2)(x - 5)$ (2) (d) Simplify fully $n^4n^7n^5$ (2)

YOUR SOLUTION

05 Rearrange formula

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = \sqrt{\frac{x+1}{x-4}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q16

Rearrange formula

$$y = \sqrt{\frac{x+1}{x-4}}$$

Square both sides

$$y^2 = \frac{x+1}{x-4}$$

Finish: $x = \frac{4y^2+1}{y^2-1}$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Square both sides

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rearranging Formulas question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = \sqrt{\frac{x+1}{x-4}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q16

Rearrange formula

$$y = \sqrt{\frac{x+1}{x-4}}$$

Square both sides

$$y^2 = \frac{x+1}{x-4}$$

Finish: $x = \frac{4y^2+1}{y^2-1}$

FORMULA BANK

Undo operations around the subject, or multiply away fractions first.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Make the subject visible
"work out", "show", "hence", a familiar exam phrase	Clear fractions first
"work out", "show", "hence", a familiar exam phrase	Undo operations in order
"work out", "show", "hence", a familiar exam phrase	Factor out the subject
"work out", "show", "hence", a familiar exam phrase	Rearrange formula
"work out", "show", "hence", a familiar exam phrase	Square both sides
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.